

University of Leeds
School of Electronic and Electrical Engineering

ELEC5870M Interim Report

**Fabrication of Enhancement Mode HEMT Devices via Gate
Recession**

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Abstract

Abstract goes here

Acknowledgements

Abbreviations

2DEG	Two-dimensional electron gas
HEMT	High-electron-mobility transistor

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CHAPTER 1

Introduction

Modern communication, sensing, and microwave systems impose increasingly demanding requirements on the semiconductor devices that form their front-end signal paths. Higher operating frequencies, tighter noise constraints, and strict linearity targets continue to push transistor technologies beyond the capabilities of many silicon-based platforms. [1] Although silicon processes have made notable progress in recent years, they often fall short in providing the necessary electron mobility, noise performance, or high-frequency gain required for performance-critical microwave and millimetre-wave applications. (e.g. silicon's $1200 \text{ cm}^2/\text{Vs}$ compared to GaAs's $6500 \text{ cm}^2/\text{Vs}$ in transistors for MMICs) [1]

III–V High–Electron–Mobility Transistors (HEMTs) play a central role in addressing these challenges. Their heterostructure–based design creates a highly conductive channel at the interface between two semiconductor layers with different bandgaps. At this interface a two–dimensional electron gas (2DEG) forms, supporting exceptionally high carrier mobility. The resulting strong transconductance, stable high–frequency behaviour and low–noise performance have firmly established HEMTs as core devices throughout microwave and millimetre–wave circuit design. [2]

Research activity within the field has increasingly shifted toward gallium nitride (GaN) HEMTs, driven by the material properties enabled by its wide bandgap. GaN supports high breakdown voltages, strong power–handling capability and robust operation under large electric fields, making it the preferred choice for many high–power and high–linearity applications [3]. Despite this growing dominance, gallium arsenide (GaAs) HEMTs remain one of the most mature and well characterised III–V transistor platforms. Their stability, extensive documentation and long industrial history continue to make them an invaluable system for studying the relationships between heterostructure design, device architecture and electrical behaviour. [4, 5]

For the purposes of this project, GaAs presents clear practical and technical advantages. The Leeds Nanotechnology Cleanroom maintains established, reliable processes for GaAs/AlGaAs heterostructures, and suitable wafers are readily available with reproducible epitaxial quality [6].

Standard GaAs/AlGaAs HEMTs are inherently depletion-mode devices (d-mode); they naturally conduct current at zero gate bias and require a negative voltage to switch off. [2] This 'normally-on' behaviour is often undesirable as it necessitates complex negative polarity power supplies [7] and complicates the design of direct-coupled logic circuits. [8] Consequently, there is significant interest in developing enhancement-mode devices (e-mode) that are 'normally-off' and offer reduced power consumption.

One common approach to achieving e-mode operation is to recess the gate into the AlGaAs donor layer, thereby reducing the amount of charge carriers available to form the channel and shifting the threshold voltage in the positive direction. [9] However, achieving e-mode operation through gate recessing is inherently non-trivial, as it requires precise and repeatable removal of only a few nanometres of material, with small variations in etch depth leading to significant changes in threshold voltage and device behaviour. [10]

The primary objective of this project is to investigate the relationship between gate recess depth and threshold voltage in GaAs HEMTs, with a focus on understanding the practical limitations and repeatability of the fabrication process. This will be achieved through the development of a gate recession process in which the gate contact is recessed into the AlGaAs donor layer to shift the threshold voltage into the positive region.

To provide a rigorous comparison between d-mode and e-mode devices, this interim report first focuses on the fabrication and characterisation of a series of non-recessed d-mode devices, achieved in first semester. (Chapters 3 and 4). These standard d-mode devices serve as a control to validate the HEMT fabrication process and establish reference parameters throughout this project. Finally, Chapter 5 outlines the theory and proposed process strategy required to implement the precise donor layer etch needed to achieve e-mode operation in the second phase of this work.

CHAPTER 2

Technical Background

2.1 Overview of GaAs/AlGaAs HEMTs

In GaAs/AlGaAs high-electron-mobility transistors (HEMTs), the interface between the GaAs channel layer and the overlying AlGaAs barrier produces a discontinuity in the conduction band due to the differing bandgaps of the two materials. This offset creates a potential well on the GaAs side of the junction, as illustrated in Figure 2.1, which confines electrons to the heterointerface and enables the formation of a two-dimensional electron gas (2DEG).

Electrons supplied by silicon donors in the modulation-doped AlGaAs layer transfer into this potential well and accumulate to form the 2DEG, which serves as the conducting channel of the device. The spatial separation of carriers from ionised dopants results in exceptionally high electron mobility, underpinning the strong transconductance, low-noise performance, and high-frequency capability that distinguish HEMTs from bulk-channel field-effect transistors [11].

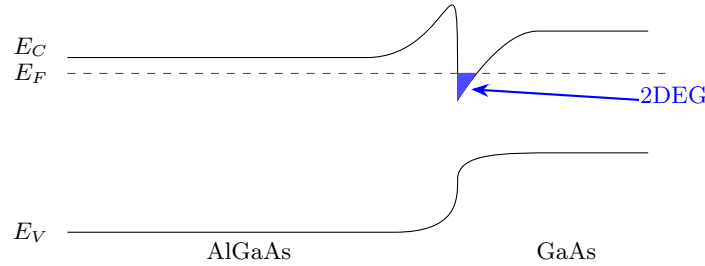


Figure 2.1: Band Structure Diagram of the AlGaAs/GaAs interface demonstrating the resulting Two-Dimensional Electron Gas (2DEG) formed in a Potential Well.

2.2 The GaAs/AlGaAs Heterostructure

The devices investigated in this project employ a conventional GaAs/AlGaAs heterostructure in which a sequence of epitaxial layers is engineered to create and control a 2DEG. A schematic cross-section of the structure is shown in Figure 2.2, highlighting the cap, donor, spacer, and channel layers whose thicknesses and compositions govern the electrostatic control and transport behaviour of the device.

At the surface of the structure lies a 10 nm thick, highly doped GaAs cap layer. This cap provides a chemically stable surface for forming the Schottky gate contact whilst protecting the underlying AlGaAs. Its high conductivity reduces series resistance in the access regions adjacent to the source and drain contacts, improving carrier injection into the channel.

Beneath the cap layer is a 40 nm thick silicon-doped AlGaAs donor layer. Silicon donors

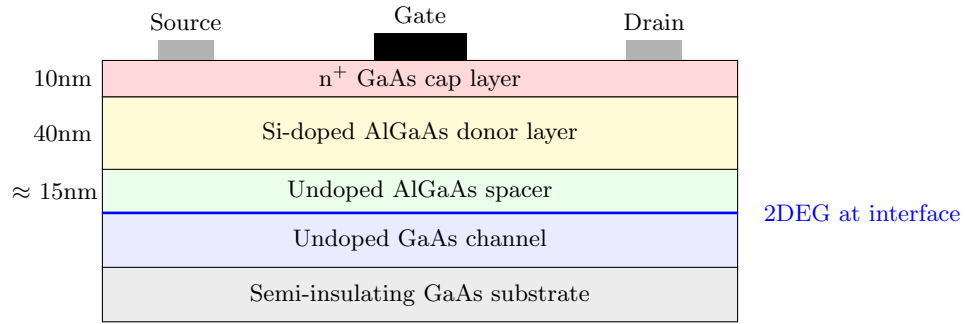


Figure 2.2: Cross-section of the GaAs/AlGaAs HEMT used in this project (not to scale).

act as electron suppliers, with carriers transferring into the GaAs channel as a result of the conduction band discontinuity at the heterointerface. This modulation-doping strategy spatially separates the ionised dopants from the conducting channel. Both the donor concentration and its thickness strongly influence the resulting sheet carrier density of the 2DEG and therefore the threshold voltage and maximum drain current of the device.

A 15 nm thick undoped AlGaAs spacer layer separates the donor layer from the GaAs channel. This spacer suppresses Coulomb scattering by physically isolating the 2DEG from ionised impurities, enabling exceptionally high electron mobility [12]. The benefit is particularly pronounced in HEMT devices operating at cryogenic temperatures, where phonon scattering is reduced and impurity scattering would otherwise dominate transport behaviour [13]. Increasing the spacer thickness improves mobility but weakens electrostatic coupling between the gate and the channel; this trade-off is a fundamental consideration in heterostructure design.

In this project, the thickness of the doped AlGaAs donor layer beneath the gate represents the primary controllable parameter through which the threshold voltage may be engineered. The following section therefore examines how gate bias modulates channel charge and determines the operational mode of the device.

2.3 Gate Control of Channel Charge and Threshold Voltage

The electrical behaviour of the GaAs/AlGaAs HEMT is governed by the field-effect interaction between the Schottky gate and the 2DEG. When a metal gate is deposited onto the semiconductor surface, a depletion region forms beneath it; the depth of this region is modulated by the applied gate voltage (V_G). This modulation controls the sheet carrier density (n_s) in the channel and, consequently, the drain current [14].

2.3.1 Charge Distribution and Effective Gate Separation

In a simple charge-control description of GaAs/AlGaAs HEMTs, the 2DEG is treated as an infinitesimally thin sheet of charge located at the GaAs/AlGaAs heterointerface, analogous to a parallel-plate capacitor; with the gate and channel acting as plates and the AlGaAs donor layer acting as a 'dielectric'. [15] This approximation is adequate for devices with relatively thick barrier layers. As the gate is recessed and the gate-to-channel separation is reduced towards the nanometre scale required for enhancement-mode operation, however, the assumption of a

sheet-like charge distribution becomes increasingly inaccurate.

In practice, electrons forming the 2DEG occupy a quantised triangular potential well on the GaAs side of the heterointerface, resulting in a charge distribution of finite thickness. The centre of this distribution is displaced from the interface, increasing the effective electrostatic separation between the gate and the channel.

Lee *et al.* account for this behaviour by introducing an effective gate-to-channel separation of $d + \Delta d$, where Δd represents a quantum-mechanical correction associated with the finite spatial extent of the electron wavefunction. For GaAs/AlGaAs HEMTs, Δd is of order 80 Å [15]. This correction may be incorporated into the transconductance parameter β , which relates changes in gate voltage to changes in drain current through the gate-controlled channel capacitance:

$$\beta = \frac{\epsilon \mu W}{(d + \Delta d)L} \quad (2.1)$$

ϵ : Permittivity of the barrier; μ : Electron mobility; W, L : Gate width and length; d : Physical gate-to-channel separation; Δd : Effective charge-centroid offset.

For the depletion-mode devices fabricated in Semester 1, the total electrostatic separation between the gate and the 2DEG is sufficiently large that this quantum correction represents a small fraction of the overall spacing. As the donor layer is thinned towards the dimensions required for enhancement-mode operation, the correction becomes a more significant contribution. Its relevance in this project lies in informing the practical recess depth required to shift the threshold voltage to positive values, and an important consideration to the transconductance of the HEMT.

2.3.2 Threshold Voltage and Mode Determination

The distinction between depletion-mode (D-mode) and enhancement-mode (E-mode) operation is determined by the balance between the built-in potential of the Schottky gate and the thickness of the doped AlGaAs donor layer. Within the analytical framework presented by Lee *et al.* [15], the threshold voltage (V_{off}) may be expressed as

$$V_{off} = \phi_b - \Delta E_c - \frac{q N_d d_d^2}{2\epsilon} \quad (2.2)$$

ϕ_b : Schottky barrier height; ΔE_c : Conduction band offset; q : Elementary charge; N_d : Donor concentration; d_d : Donor thickness; ϵ : Permittivity of the AlGaAs barrier.

This quadratic dependence of V_{off} on donor layer thickness highlights the strong sensitivity of the threshold voltage to small variations in etched depth. This relationship directly underpins the two operational modes explored in this project.

Depletion Mode (Semester 1): The baseline devices fabricated in Semester 1 employed the full epitaxial structure beneath the gate. Under these conditions, the zero-bias depletion region does not extend to the heterointerface, leaving the 2DEG populated ($V_{off} < 0$) and the device normally on, such that a negative gate voltage is required to pinch off the channel [5].

Enhancement Mode (Semester 2 Objective): To achieve normally-off behaviour ($V_{off} > 0$), the donor layer must be thinned until the depletion region fully intersects the 2DEG at zero gate bias. For a donor concentration of $1 \times 10^{18} \text{ cm}^{-3}$, the critical donor thickness is approximately 25 nm [15]; this finding by Lee *et al.* serves as a well-grounded guide for the similar GaAs/AlGaAs material structure employed in this project. Recessing the gate to this depth shifts the threshold voltage, such that a positive gate bias is required to re-populate the channel and initiate conduction [9].

Together, these relationships demonstrate that achieving enhancement-mode operation in GaAs/AlGaAs HEMTs depends critically on precise control of the donor layer thickness beneath the gate, forming the physical basis for the fabrication strategy developed in Chapter 5.

CHAPTER 3

Fabrication of Depletion-Mode HEMT Devices

3.1 Overview of the Fabrication Workflow

The depletion-mode HEMTs characterised in this report were fabricated using the standard III-V device process established within the Leeds Nanotechnology Cleanroom. This workflow follows the routine sequence used for GaAs/AlGaAs transistors in this facility: it comprises of 3 main stages: mesa definition, ohmic contact formation, and gate formation. This well-established procedure was followed without modification and carried out over four full-day laboratory sessions. The devices produced through this workflow provide a necessary baseline; they allow for a reliable comparison against which the behaviour of e-mode devices with different structures and dimensions can be evaluated later in this project.

Day 1 - Mesa Formation	Cleaning	Lithography	Wet Etching	Cleaning		
Day 2 - Ohmic Contacts	Cleaning	Lithography	Metal Deposition	Metal Lift-off	Cleaning	
Day 3 - Gate Formation	Annealing	Cleaning	Lithography	Metal Deposition	Lift-off	Cleaning
Day 4 - Electrical Testing	Microscope Inspection	Defect Removal	Electrical Characterisation			

Figure 3.1: Timeline overview of the four-day cleanroom workflow used to fabricate and characterise baseline depletion-mode GaAs/AlGaAs HEMTs.

3.2 Mesa Definition and Isolation Etch

The fabrication process began with the electrical isolation of individual device regions through mesa etching. Prior to lithography, the GaAs wafer was cleaned sequentially in acetone, isopropanol (IPA), and deionised water (DI) to remove organic residues and ensure sufficient photoresist adhesion. To further enhance the bonding interface between the substrate and the resist, a Surpass primer was spun onto the wafer. This priming step is critical for wet-etching applications: it modifies the surface tension to prevent the etchant from penetrating under the resist edges (an effect known as undercutting) which would otherwise lead to the delamination of the mask. A positive photoresist (S1803) was then spun onto the sample at 3000 rpm, soft-baked on a hot-plate at 115 °C, and subsequently patterned with exposure from the Heidelberg MLA 150 Maskless Aligner (MLA).

Following exposure, the sample was immersed in a caustic Microposit developer solution for 45 seconds, followed by a thorough 2 stage DI water rinse and a nitrogen blow-dry to reveal the mesa patterns. The mesa pattern was then transferred into the wafer using a sulphuric acid,

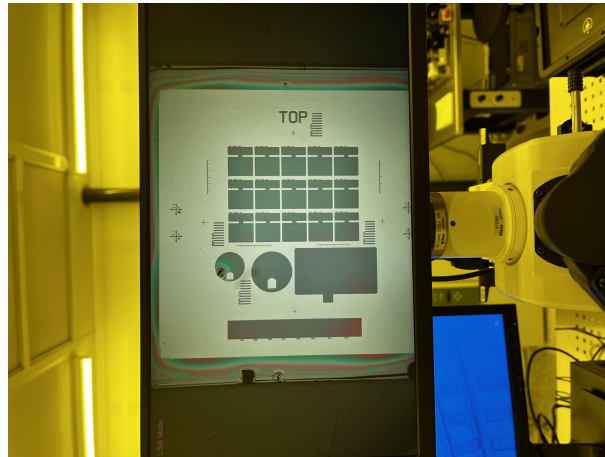


Figure 3.2: Mesa lithography mask used for defining isolated device regions (Post-Development).

hydrogen peroxide, and water wet-etch solution ($H_2SO_4 : H_2O_2 : H_2O$ in a 1:8:160 ratio). This etchant is highly suited to GaAs/AlGaAs mesa formation: it produces a controllable, moderately fast isotropic etch that removes the active layers uniformly. The rate is sufficiently slow to allow for an accurate timed etch of a shallow ≈ 200 nm recess, yet it remains fast enough to be practical for routine device isolation.

To calibrate the process, an initial 30 s pilot etch was performed; subsequent measurement using an Alpha-Step profilometer revealed an etch depth of 114 nm, corresponding to an initial etch rate of 3.8 nm s^{-1} . To reach the target depth of 200 nm, a further 86 nm of material required removal. Assuming a constant etch rate, a second etch duration of 22 s was calculated and performed; however, the final measured depth reached only 177 nm. This indicates that the second etch only removed an additional 63 nm at a reduced effective rate of approximately 2.9 nm s^{-1} . While a total depth of 177 nm is sufficient for device isolation (as it successfully penetrates the active GaAs/AlGaAs heterostructure layers), the discrepancy in rates highlights the intrinsic sensitivity of the wet-etch chemistry.

A theory for the observed reduction in the mean etch rate during the second step is attributed to the formation of a native oxide layer on the GaAs surface during the interval in which the sample was removed for profilometry. This oxide layer creates an "incubation period" at the start of the second etch: sample etching is delayed while the etchant penetrates the oxide before reaching the underlying semiconductor. Because the second etch (22 s) was shorter than the first (30 s), this non-productive period represented a larger percentage of the total etch time; consequently, the average rate for that specific step was significantly lowered.

[PROVIDE LITERATURE BACKUP TO THIS THEORY SUGGEST MITIGATION]

3.3 Ohmic Contact Formation

The formation of low-resistance Ohmic contacts is a critical prerequisite for accessing the two-dimensional electron gas (2DEG) within the heterostructure. When metal is deposited directly onto the GaAs cap, the interface naturally forms a Schottky barrier due to Fermi-level pinning at the surface states: this creates a rectifying junction that restricts carrier injection. To overcome

this, a specific metallisation scheme and thermal processing step are required to facilitate field-emission (tunnelling) conduction. The objective is to create a heavily doped n^+ region beneath the contact pads that extends down towards the channel; this high doping density significantly narrows the depletion width of the barrier, allowing electrons to tunnel freely between the metal and the semiconductor with linear $I - V$ characteristics.

To define these contacts, a dual-layer photoresist stack comprising Lift-Off-Resist (LOR) beneath S1803 was employed. A single photoresist layer typically develops near-vertical sidewalls: this causes the evaporated metal to form a continuous film across both the wafer surface and the photoresist edges, making lift-off unreliable and often leaving metallic "fences" around the contact perimeter. The dual-layer stack mitigates this by exploiting the higher developer solubility of the LOR base layer. During development, the LOR recedes laterally more than the S1803 above it, producing a distinct undercut profile. This physical overhang breaks the continuity of the deposited metal film, ensuring that the metal on top of the photoresist is electrically and mechanically separated from the device contacts. Consequently, solvent-based lift-off yields sharply defined openings free of edge defects.

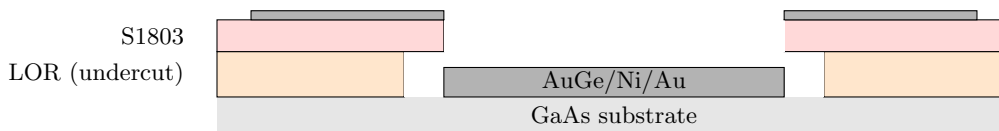


Figure 3.3: Dual-layer LOR/S1803 resist stack after development showing the wider LOR opening that creates an undercut for clean metal lift-off.

Following lithography, the Ohmic contacts were formed by thermally evaporating a AuGeNi eutectic pellet. This alloy is commonly used for n-type GaAs Ohmic contacts because each element plays a specific role during annealing. Germanium diffuses into the GaAs and acts as a donor, creating a highly doped region beneath the contact. Nickel improves the uniformity and adhesion of the metal film and supports the diffusion process, whilst gold provides a low-resistance conduction path between the contact pad and the semiconductor.

To activate the contacts, the sample underwent Rapid Thermal Annealing (RTA). It is during this thermal cycle that the transition from rectifying to Ohmic behaviour occurs. The heat drives the Germanium atoms to diffuse through the AlGaAs layers and into the GaAs lattice, locally creating the n^+ doped region. This effectively "thins" the potential barrier at the interface, establishing a low-resistance tunnelling path that connects the drain and source electrically to the 2DEG.

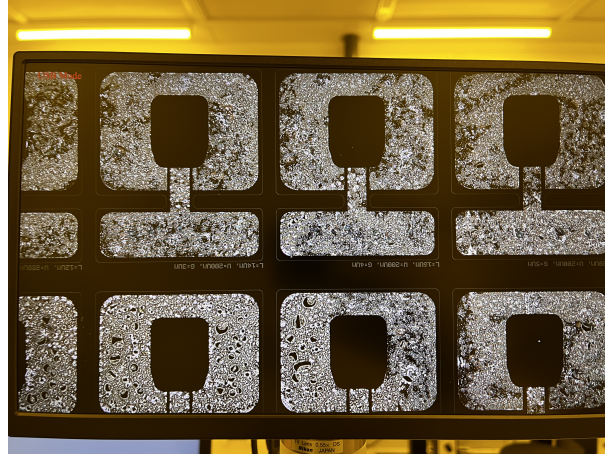


Figure 3.4: Ohmic contacts under x5 microscopy following lift-off.

3.4 Gate Formation and Final Inspection

With the Ohmic contacts established and annealed, a third photolithography step was performed to define the Schottky gate region. As with the previous stage, a dual-layer resist stack was utilised to ensure a clean lift-off. The gate electrode was aligned centrally between the source and drain contacts; its placement is critical, as it provides the electrostatic control required to modulate the carrier density within the channel.

Following development, a Ti/Au stack was evaporated to form the gate electrode. In this configuration, the Titanium (Ti) layer serves as a vital adhesion layer to the semiconductor surface, while the thick Gold (Au) layer provides a stable Schottky barrier and minimises gate resistance. Unlike the Ohmic contacts, the gate metallisation is not annealed; preserving the Schottky barrier height is essential for minimising leakage current and maintaining effective depletion control over the channel.

After metallisation, the resist was removed using solvent lift-off to reveal the final device structure. A small-number of lift-off defects (primarily isolated metallic particles) were observed under the microscope and subsequently removed using brief, gentle ultrasonic cleaning. The removal of these artefacts is essential for device reliability: they can otherwise cause electrical short circuits, particularly around the sensitive gate edges.

The completed devices were examined using optical microscopy to ensure process integrity. The mesas exhibited clean, well-defined sidewalls; the Ohmic pads were properly positioned; and the gate electrodes showed sharp, uninterrupted edges with no evidence of bridging or residual flakes. Overall, the fabrication process yielded a high-quality array of depletion-mode HEMTs suitable for subsequent electrical characterisation.

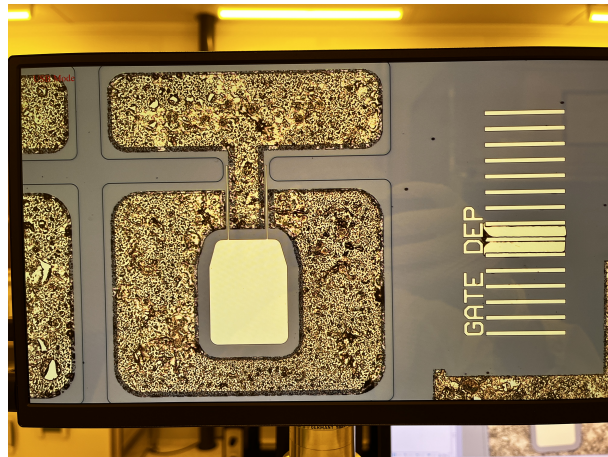


Figure 3.5: Gate contacts after lift-off.

CHAPTER 4

Electrical Characterisation and Critical Analysis

- 4.1 Measurement Setup
- 4.2 Output Characteristics (ID–VD)
- 4.3 Transfer Characteristics (ID–VG)
- 4.4 Gate Leakage Behaviour
- 4.5 Threshold-Voltage Extraction
- 4.6 Variability, Non-Idealities and Interpretation
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CHAPTER 5

Exploration of Enhancement-Mode HEMT Devices

- 5.1 Motivation for Enhancement-Mode Operation
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- 5.5 Ideal Outcomes and Target Behaviour
- 5.6 Logic Motivation and Stretch Goal

CHAPTER 6

Reflection and Project Management

6.1 Summary of Progress

6.2 Challenges and Adaptations

6.3 Skills Developed

6.4 Lessons for Semester Two

CHAPTER 7

Semester Two Plan

7.1 Gantt Chart

7.2 Planned Technical Deliverables

7.3 Risk Mitigation Strategy

Bibliography

- [1] A. Chaturvedi, “Review of semiconductor materials for monolithic microwave millimeter wave integrated circuits,” *Materials Today: Proceedings*, vol. 37, pp. 1567–1570, 2021.
- [2] U. Mishra, P. Parikh, and Y.-F. Wu, “Algan/gan hemts-an overview of device operation and applications,” *Proceedings of the IEEE*, vol. 90, no. 6, pp. 1022–1031, 2002.
- [3] U. K. Mishra, L. Shen, T. E. Kazior, and Y.-F. Wu, “Gan-based rf power devices and amplifiers,” *Proceedings of the IEEE*, vol. 96, no. 2, pp. 287–305, 2008.
- [4] T. Mimura, “The early history of the high electron mobility transistor (hemt),” *IEEE Transactions on Microwave Theory and Techniques*, vol. 50, no. 3, pp. 780–782, 2002.
- [5] T. Mimura, S. Hiyamizu, T. Fujii, and K. Nanbu, “A new field-effect transistor with selectively doped gaas/n-alxga1-xas heterojunctions,” *Japanese Journal of Applied Physics*, vol. 19, p. L225, may 1980. The seminal paper for the first GaAs HEMT devices.
- [6] The University of Leeds, “Leeds nanotechnology cleanroom process gallery.” Available at: <https://cleanroom.leeds.ac.uk/>.
- [7] STMicroelectronics, *E-mode GaN technology: tips for best driving*, 2021. Explains why D-mode is undesirable for power converters due to ”shoot-through” risks at startup and the complexity of negative gate-drive circuitry.
- [8] X. Y. Q. S. et al, “Monolithically integrated enhancement/depletion-mode algan/gan hemt d flip-flop using fluorine plasma treatment,” *Journal of Semiconductors*, vol. 32, p. 065001, jun 2011.
- [9] W. Saito, Y. Takada, M. Kuraguchi, K. Tsuda, and I. Omura, “Recessed-gate structure approach toward normally off high-voltage algan/gan hemt for power electronics applications,” *IEEE Transactions on Electron Devices*, vol. 53, no. 2, pp. 356–362, 2006.
- [10] W. Lanford, T. Tanaka, Y. Otoki, and I. Adesida, “Recessed-gate enhancement-mode gan hemt with high threshold voltage,” *Electronics Letters*, vol. 41, pp. 449–450, 2005.
- [11] W. Liu, *Fundamentals of III-V Devices: HBTs, MESFETs, and HEMTs*. John Wiley and Sons, 1999.
- [12] K. Hirakawa and H. Sakaki, “Mobility of the two-dimensional electron gas at selectively doped n -type al_xga_{1-x}as/gaas heterojunctions with controlled electron concentrations,” *Phys. Rev. B*, vol. 33, pp. 8291–8303, Jun 1986.
- [13] L. Pfeiffer, K. W. West, H. L. Stormer, and K. W. Baldwin, “Electron mobilities exceeding 10⁷ cm²/V·s in modulation-doped GaAs,” *Applied Physics Letters*, vol. 55, no. 18, pp. 1888–1890, 1989. A seminal work demonstrating record mobilities at cryogenic temperatures by optimising spacer thickness to eliminate impurity scattering.

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- [14] S. M. Sze and K. K. Ng, *Physics of Semiconductor Devices*. Hoboken, NJ: John Wiley & Sons, 4th ed., 2021. The industry-standard reference for semiconductor devices physics including Schottky barrier formation and the depletion region physics in III-V devices.
 - [15] K. Lee, M. Shur, T. Drummond, and H. Morkoc, “Current—voltage and capacitance—voltage characteristics of modulation-doped field-effect transistors,” *IEEE Transactions on Electron Devices*, vol. 30, no. 3, pp. 207–212, 1983.

APPENDIX A

Additional Figures

APPENDIX B

Code and Process Flow